

Kelsey Knight

Isabella Vera

Eunji Kim

Zoe Tseng

Project title: Predicting Household Energy Burden from Urban Form and Socioeconomic Characteristics in New York City

Background: This project investigates whether urban form variables - building age, density, floor area ratio, gross building area, tree canopy coverage, and transit proximity - combined with socioeconomic characteristics such as household income, tenure, and household size, can predict household energy burden at the census tract level across New York City. Energy burden is defined as the share of household income spent on energy costs. Roughly 25% of NYC residents spend more than 6% of their income on utilities (NYC Comptroller, 2025), a threshold considered a high burden. This project applies supervised regression and unsupervised clustering to identify which features most strongly predict burden and to group neighborhoods into typologies.

Project Goal: If the full model (with urban form variables) outperforms the demographic-only baseline, this suggests that neighborhood physical conditions contribute to energy burden through mechanisms beyond socio-demographic indicator, with implications for more place-based policy interventions (rather than solely income/demographic based).

Machine Learning:

- Linear Regression: Sociodemographic indicators to establish what component of energy burden can be attributed to census related data points. Regression results will be used for later comparison to the RF results.
- Random Forest: different variables (building age, density, floor area ratio, gross building area, tree canopy coverage, and transit proximity) to predict energy burden
 - Follow-up full RF by running with sociodemographic features only. The change in R^2 relative to the full model quantifies the independent contribution of urban form. This comparison helps answer the research question in terms of contribution of urban form variables.
- K-means: Determine natural groupings of NYC census tracts to see if they align with actual energy burden groupings (determining mean energy burden per cluster). We will use the elbow method to select the number of clusters k , with an expected result of 4 to 6.

Datasets:

Dataset	Description	URL
DOE LEAD Tool	Modeled energy burden at census tract level; direct CSV download or API access	energy.gov/scep/slsc/lead-tool
NYC PLUTO	Parcel-level land use, building age, FAR, and gross floor area for all NYC tax lots; shapefile or CSV	https://data.cityofnewyork.us/City-Government/Primary-Land-Use-Tax-Lot-Output-PLUTO-64uk-42ks/about_data
Tree equity index	2015 census of all NYC street trees with location, species, and condition; CSV download	https://data.cityofnewyork.us/Environment/2015-Street-Tree-Census-Tree-Data/uvpi-gqnh/about_data
MTA Subway Stations / Bus stops	Point locations of all NYC subway stations and bus stops; used to compute distance-to-transit features by tract	https://data.ny.gov/Transportation/MTA-Subway-Stations/39hk-dx4f/about_data https://data.ny.gov/Transportation/MTA-Bus-Stops/2ucp-7wg5/about_data
ACS 5-Year Estimates	Tract-level income, tenure, household size, racial composition, and housing vintage from the Census Bureau	data.census.gov
CEJST	EPA energy burden indicator and disadvantaged community designations at the tract level; used for validation	https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/B6ULET

Variables

- Trees: surface temperature in census tract
 - Would need to do zipcode to census tract mapping (only cover 80% of census tract)
- Build material: energy efficiency <https://zenodo.org/records/10124952>
- PLUTO
 - Floor area ratio
 - Census tract
 - Borough
 - Building class
 - Land use
 - Year built
 - FAR

Machine Learning Project Overview

	borough	zipcode	ct2010	landuse	zonedist1	bldgclass	builtfar	yearbuilt	unitsres	unittotal
0	BX	10455	83	6	C8-3	E7	1.950000000000	2023	0	1
1	BX	10473	28	NaN	M1-1	Z9	0.010000000000	1955	0	1
2	BX	10473	90	1	R3-2	A1	0.700000000000	1960	1	1
3	BX	10473	42	8	R6	W9	0.540000000000	1960	0	0
4	BX	10463	283	5	M1-1	K1	0.350000000000	1951	0	1
5	BX	10473	4	1	R3A	B2	0.260000000000	1930	2	2
6	BX	10473	4	1	R3A	B2	0.760000000000	2004	2	2
7	BX	10473	4	1	R3A	B9	0.230000000000	1920	2	2
8	BX	10473	4	1	R3A	B3	0.460000000000	1930	2	2
9	BX	10473	4	1	R3A	A2	0.190000000000	1925	1	1

- Census data (median income, race)
- Median income: B19013_001E
- Race total: B02001_001E
- White alone: B02001_002E
- Tenure total: B25003_001E
- Renter occupied: B25003_003E
- Average household size: B25010_001E
- Housing burden total renters: B25070_001E
- Housing burdened renters 30%+: B25070_007E to B25070_010E

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Table	Topic	Columns to keep	Derived variable
B19013	Median household income	B19013_001E	Keep as-is → median_income
B25003	Tenure (owner vs. renter)	B25003_001E (total) B25003_003E (renter)	pct_renter = _003E / _001E
B25010	Average household size	B25010_001E	Keep as-is → avg_hh_size
B17001	Poverty status	B17001_001E (total) B17001_002E (below poverty)	pct_poverty = _002E / _001E
B02001	Race	B02001_001E (total) B02001_002E (white alone)	pct_nonwhite = 1 - (_002E / _001E)
B25034	Year structure built	B25034_001E (total) B25034_010E (built ≤1939)	pct_pre1940 = _010E / _001E
B25070	Gross rent as % of income	B25070_001E (total renters) B25070_007E-010E (30%+ burdened)	pct_rent_burdened = sum(_007-010E) / _001E
B25002	Occupancy status (vacancy)	B25002_001E (total units) B25002_003E (vacant)	vacancy_rate = _003E / _001E

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For NYC, you only need the 5 borough counties in New York State:

- Bronx = 005
- Kings (Brooklyn) = 047

Machine Learning Project Overview

- New York (Manhattan) = 061
- Queens = 081
- Richmond (Staten Island) = 085
- Transit proximity
 - Census tract
 - Subway stops counts (R: 0.5 miles)
 - Bus stops counts (R: 0.25 miles)
 - Subway stops: [MTA Subway Stations | State of New York](#)
 - Bus stops: [MTA Bus Stops | State of New York](#)

```
--- Mean counts by borough ---
      subway_025mi subway_05mi subway_10mi bus_025mi bus_05mi \
borough
BRONX              0.52       1.97        7.09       11.38      43.48
BROOKLYN           0.61       2.27        8.46       14.02      54.21
MANHATTAN          1.27       4.86       17.22       18.68      70.63
QUEENS             0.27       0.99        3.54        8.20      31.79
STATEN ISLAND      0.07       0.29        1.02        6.74      29.47

      bus_10mi
borough
BRONX          162.03
BROOKLYN       203.28
MANHATTAN      248.91
QUEENS         120.08
STATEN ISLAND  107.22
```

```
--- Final Check ---
Shape: (2327, 7)
No duplicate GEOIDs
No nulls
No negative values

All checks passed. Ready to save.
```

	subway_025mi	subway_05mi	subway_10mi	bus_025mi	bus_05mi	bus_10mi
count	2327.00	2327.00	2327.00	2327.00	2327.00	2327.00
mean	0.55	2.06	7.48	12.02	46.41	171.84
std	0.90	2.47	7.08	8.06	23.92	75.25
min	0.00	0.00	0.00	0.00	0.00	0.00
25%	0.00	0.00	2.00	6.00	29.00	119.00
50%	0.00	2.00	7.00	11.00	42.00	165.00
75%	1.00	3.00	11.00	17.00	62.00	222.00
max	7.00	19.00	39.00	55.00	153.00	434.00